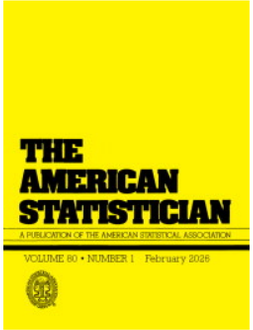

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Revisiting Bessel's Correction and the Bias-Variance Tradeoff in Variance Estimation

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Abstract

When estimating the variance from a sample, usually the so-called Bessel's correction is used, i.e., unintuitively each term is weighted by the sample size *n* minus one. Although this is an unbiased estimator, it does not necessarily yield the best accuracy in terms of bias-variance tradeoff. To this end, we conducted bibliometric work and observe that many statistics textbooks recommend Bessel's correction for somewhat spurious reasons. Furthermore, we address the bias-variance tradeoff in variance estimation from a theoretical perspective and show that usually the uncorrected version of the sample variance is a better choice. We back this up with a simulation study that can be conducted in a classroom setting, where students can learn about unbiasedness, bias-variance tradeoff, conducting simulation studies, and estimation in general from an easy example.

Keywords: Variance; Estimation; Simulation; Teaching

JEL classification: A20;C13;

1 Introduction

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample of the univariate random variable *X*, where *X* has distribution function $F(x)$ and we are faced with the problem to estimate the variance of *X* from the sample. Now pause to think for a moment. How would you do this, or, which formula would you use? A large amount of the statistics literature suggests one of the following two well-known formulas, which we will call *sample variance* S_n^2 and *corrected sample variance* S_{n-1}^2, respectively, where the subscript refers to the divisor (and not to the sample size). Let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ denote the sample mean. Then

$$S_n^2 := \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2, \quad (1)$$

$$S_{n-1}^2 := \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \quad (2)$$

are consistent estimators of the variance σ_X^2 of *X*. Let us also assume that the fourth moment of *X* exists, so that both estimators have finite variance. Using the mean squared error (MSE) as a measure of the accuracy of a point estimate, we can show that

$$\mathrm{MSE}\left(S_n^2\right) < \mathrm{MSE}\left(S_{n-1}^2\right) \quad (3)$$

holds in many cases (see Section 3 and see Table 2 for examples).

From the decomposition

$$\mathrm{MSE} = \mathrm{Variance} + \mathrm{Bias}^2$$

(i.e., the *bias-variance tradeoff*), we understand that two components, namely variance and bias, influence the relation in Formula (3). For the variance component we have

$$\mathrm{Var}\left(S_{n-1}^2\right) > \mathrm{Var}\left(S_n^2\right)$$

which follows directly from

$$S_{n-1}^2 = \frac{n}{n-1} S_n^2.$$

Since S_n^2, the estimator with lower variance, is biased, while S_{n-1}^2 is unbiased, it is to be expected that the interplay of sample size and the underlying distribution $F(x)$ (both influencing

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bias and variance) leads to settings in which either estimator has the smaller MSE. That is indeed the case: For normally distributed *X*, we will later derive that for any sample size the uncorrected estimator S_n^2 exhibits a smaller MSE than S_{n-1}^2. Surprisingly, this holds also for all distributions with a kurtosis larger than 1.5, which includes most of the commonly-studied distributions, making S_n^2 an attractive choice.

If you selected Formula (2), the corrected estimator for the problem above, rest assured that you are not alone. The factor $\frac{1}{n-1}$, known as Bessel's correction, dominates statistical education. Many textbooks barely acknowledge the uncorrected sample variance S_n^2, and even ChatGPT-4o presents the corrected version as the sole answer to the prompt, "How to calculate the sample variance?". Among 143 examined books, 125 introduce S_{n-1}^2, and 98 of them do not even mention S_n^2 as an estimator. This preference is also reflected in students' understanding. When we asked Bachelor's and Master's students from various disciplines at different German universities how they estimate variance, nearly all recommended using S_{n-1}^2. Some even argued that one should never use S_n^2 *since it is wrong and must be corrected*.

In a highly enjoyable article, Jeffrey S. Rosenthal discusses the question of whether *n* or $n-1$ should be used as the denominator when teaching students the concept of sample variance (Rosenthal 2015). According to Rosenthal, common responses to students questioning the use of $n-1$ in S_{n-1}^2 include "a confusing and premature explanation of unbiased estimators", deferring the explanation by saying it "will be explained later", arguing that "if *n* is large then "it doesn't really matter"", or asserting that for "$n=1$ (…) the answer should be undefined, not zero". Finally, the ultimate justification often presented is that "everyone knows that unbiased estimators are *so* important", making all other concerns fade into insignificance. All this may contribute to the one-sided or merely existing explanations of this topic found in some statistics textbooks, with some authors even advocating against teaching sample variance in introductory courses altogether (Rumsey 2009).

Given the widespread confusion among students and the challenges in teaching variance estimation, this paper makes the following three key contributions: (1) We investigate the prevalence of both variance estimators in the literature and statistical software. (2) We demonstrate that, in many relevant cases, the uncorrected estimator is superior in terms of MSE and explicitly identify the cases where this does not hold. (3) We illustrate how this topic can be made pedagogically accessible through a simulation study.

The paper is structured as follows: In Section 2, we examine the status quo of variance estimation in textbooks and statistical software. This bibliometric analysis provides insight into how variance estimators are presented across disciplines. Without aiming to offend, we highlight some of the most simplified explanations that may leave learners in the dark. For the sake of piety, we do not individually cite the examples given, but list all considered sources in the reference list.

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Next, in Section 3, we compile and present theoretical results on variance estimation, where we focus on estimators of the form

$$S^2_{k,M,n} := \frac{1}{k}\sum_{i=1}^{n}\left(X_i - M\right)^2, \quad (4)$$

where M is either $\bar{X}$ or $E[X]$. We analyze different distributional properties of these estimators and determine the MSE-optimal choice of k. Additionally, we briefly extend our analysis to covariance estimation.

In Section 4, we present variance estimation as both an educational introduction to statistical computer simulations and a bridge between the two concepts. We analyze key findings of this paper through a simulation study, extending our investigation to accuracy metrics (MAE) and variance measures (standard deviation) that are challenging to derive analytically. We guide the reader through the structure of the simulation study and provide a well commented jupyter notebook in an online repository.

Finally, the concluding section summarizes our findings and offers final remarks.

2 Variance Estimation in the Literature

In this section, we present some bibliometric work on variance estimation. Additionally, we showcase Wikipedia's take on this as well as how the estimation is implemented in major software.

2.1 Estimation in Analyzed Statistics Textbooks

As a precaution, we mention here, that the bibliometric part is a pure and surely subjective description of the books we considered. It neither claims to be representative of a much larger choice of books nor did we try to proportionally represent all disciplines with statistics books. A listing can be found in the appendix.

We evaluated 143 educational statistics books in English and German, of which 92 were written in English and 51 in German. They were published between 1962 and 2018, but only 10 of them were published before 1997. More detailed, 20 were published from 1962-1999, 52 were published from 2000-2009 and 71 where published from 2010-2018. We categorized books by their target group, differing between general (76), business & economics (34), mathematics (10) and others (23). The group of 'others' contains various fields ranging from physics and engineering over law and sport science up to social studies and psychology.

Of these 143 books, there were 14, which only considered solely the uncorrected sample variance, most of which were contributed by the English literature. This is in accordance with our general impression, that the uncorrected sample variance is rarely known by German students. The vast majority of 98 books presented only the corrected sample variance. Here the ratio of German and English literature is similar to the overall ratio of German and English

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literature. Only 27 books introduced both types of variance estimation and the remaining four books did not mention variance estimation at all.

Most authors (112) make an implicit recommendation on which type of estimator should be used by presenting either S_{n-1}^2 or S_n^2. If both estimators are presented, most authors generally recommended the corrected sample variance. Besides arguments including the bias, some simply state that S_{n-1}^2 *is a better estimator* or *the best estimator*. Another author purely claimed S_n^2 *underestimates the variance* without any further discussion. Last, one book recommended an *empirical rule of thumb for everyday use: Reducing the division factor increases the variance, but this difference is considered acceptable already from n>30.* Only one source recommended the uncorrected sample variance, because the corrected version *is not consistent with the original definition of variance* and *unbiasedness is not a desirable property in many situations*, but did not provide more detailed explanations of both statements.

Of course, a deeper discussion of the variance and bias of an estimator would be eased by introducing the mean squared error (MSE) as a measure. In the present sample of books, there was not a single one from the field of mathematics in which the MSE was introduced. Of the other three fields, economics, general, and others, about 12% of the literature goes as far as explaining the MSE conceptually in the context of statistical testing. Not all books that explain the MSE in the context of evaluation of estimators explicitly introduce different estimators of the sample variance. The proportions are shown in Figure 1. However, the MSE concept allows the skilled reader to deduce other possible correction terms.

2.2 Estimation in Wikipedia

Nowadays, another widely used source is the web page wikipedia.org. There, the uncorrected sample variance is introduced by applying the descriptive variance of a finite population to a sample. The subsequent computation of its expected value clearly reveals the bias. This motivates the introduction of the unbiased, corrected sample variance with the Bessel's correction factor. However, in the introduction of the corresponding chapter, the concepts of mean squared error, shrinkage estimation and the resulting estimators are mentioned. The authors note, that both, corrected and uncorrected sample variance may be referred to as sample variance (Last edit on 10 June 2023).

Following the Wikipedia link to Bessel's correction, the reader gets profound insights into the ideas of bias reduction but also learns about negative features like MSE increase and how to avoid these. It is clearly pointed out, that bias and MSE can not be minimized simultaneously (Last edit on 8 March 2023).

2.3 Variance Functions Implemented in Software

When making use of variance functions implemented in standard computer software, one should be aware which of the formulas (1) and (2) is actually used. Note that this is especially important since the reason to use the variance function might not be necessarily estimation, but can be purely descriptive. So having data on the full population or to describe the variability of the data

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at hand, we would naturally want to calculate the descriptive variance of the population, which coincides with the uncorrected estimator S_n^2 in Equation (1).

We have evaluated the most obvious built-in functions one would use for this task of various statistical computer programs and programming languages. Almost all considered environments use the corrected sample variance as standard case. Those were Stata, STATISTICA, SPSS, Maple, Matlab, Python, R, SAS, SQL, Wolfram, gretl, PPSP, Orange, WEKA, KNIME, RapidMiner and EViews (see Table 1). Also in Java with the class *variance* the corrected sample variance is implemented. In some environments, a different function or an optional parameter may yield the uncorrected version. For example, the base Python module *statistics* includes the function *variance*, which returns the corrected sample variance. The uncorrected estimator is implemented in the function *pvariance*. Using the more popular *var()* function of the *numpy* package, calculates the uncorrected version but comes with a parameter to flexibly adjust the degrees of freedom similar to Equation (4). Analogously in Matlab, the command *var* computes the corrected version, but setting the optional parameter *w* equal to 1 returns the uncorrected version. For SQL the functions *VARIANCE* or *VAR_SAMP* correspond to the corrected version. For the uncorrected version the command *VAR_POP* can be used. In Microsoft Excel the functions *VAR.S* for corrected sample and *VAR.P* for uncorrected sample variance are implemented. The old name *VAR* of *VAR.S* has been replaced for the sake of clarity.

3 On Point Estimates of Variance

We now consider well-known and not-so-well-known results on the mean squared error for variance estimators of the type $S_{k,M,n}^2 := \frac{1}{k}\sum_{i=1}^{n}(X_i - M)^2$, where M is either the sample mean $\bar{X}$ or the population mean $\mu = E[X]$. We omit the subscripts M and n when we refer to the sample mean, so that $S_k^2 = S_{k,\bar{X},n}^2$; in particular, S_n^2 and S_{n-1}^2 recover the two estimators introduced above. We explicitly do not discuss a broader scope of shrinkage estimation, Bayesian estimation, or robust estimation. For these topics we refer to works of Ledoit and Wolf (2012); Muirhead (1987), Leonard and Hsu (1992); Daniels and Kass (1999), Fan, Fan, and Lv (2008); Rousseeuw and Driessen (1999); Avella-Medina, Battey, Fan, and Li (2018); Cai and Liu (2011); Rothman, Levina, and Zhu (2009); Xue, Ma, and Zou (2012).

In the following, let $\mu_j = \mathbb{E}[(X-\mu)^j]$ denote the *j*-th central moment of the population and let $\mu_4 < \infty$. Note that for the variance $\sigma^2 = \mu_2$ of the population we use both symbols for the sake of readability. We begin with a lemma stating the MSE for the estimator S_k^2.

Lemma 1 .

For $1 \le k \le n$, *the mean squared error* $\mathrm{MSE}(S_k^2) = E[(S_k^2 - \sigma^2)^2]$ *is given by*

$$\mathrm{MSE}(S_k^2) = \left(\frac{n-1-k}{k}\right)^2 \mu_2^2 + \left(\frac{n-1}{k}\right)^2 \frac{1}{n}\left[\mu_4 - \frac{n-3}{n-1}\mu_2^2\right] \quad (5)$$

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$$[0.2cm] = \mu_2^2 \left[\left(\frac{n-1-k}{k} \right)^2 + \left(\frac{n-1}{k} \right)^2 \frac{1}{n} \left(\frac{\mu_4}{\mu_2^2} - \frac{n-3}{n-1} \right) \right]. \quad (6)$$

Proof.

We recall the variance-bias decomposition of the mean squared error $\mathrm{MSE}(S_k^2) = \mathrm{Bias}(S_k^2)^2 + \mathrm{Var}(S_k^2)$. The expressions for

$$\mathrm{Bias}(S_k^2) = \frac{n-1-k}{k} \mu_2,$$

and

$$\mathrm{Var}(S_k^2) = \left(\frac{n-1}{k} \right)^2 \frac{1}{n} \left(\mu_4 - \frac{n-3}{n-1} \mu_2^2 \right)$$

follow from the calculations in Appendix 7, based on a classical moment decomposition of $\sum_{i=1}^{n} (X_i - \bar{X})^2$ (see, e.g., Mood, Graybill, and Boes (1974a); the authors state both expressions for the case $k = n-1$). Inserting these results into the bias-variance decomposition of the MSE yields (5)–(6). ▪

We see that for fixed values of μ_2 and *n*, the MSE of the estimators is a linear increasing function in the kurtosis μ_4 / μ_2^2 of the population distribution. These formulas hold for all population distributions with finite fourth moment. Note that formulas for the variance or the MSE of the variance estimators in textbooks, where the fourth moment is expressed by multiples of the second moment, refer to certain families of population distributions, e.g., normal distributions, and are less general. Be aware that there is no natural or best accuracy measure for the estimation of variance as discussed in Casella and Berger (2002a). However, we focus on the MSE here, as it is analytically tractable, unlike the Mean Absolute Error (MAE). MSE is widely used in practice and, to the best of our knowledge, covered in most basic university statistics courses. The MAE will be examined through simulation in Section 4.

3.1 Special Results for the Sample Variances S_n^2 and S_{n-1}^2

Let us now take a closer look at some results for the two estimators introduced at the beginning, i.e., Formula (1) and Formula (2). Clearly, the bias term vanishes for $k = n-1$ and in that case the MSE reduces to the variance of $S_{n-1}^2 = S_{n-1,\bar{X},n}^2$. Thus, the biased but lower-variance estimator $S_n^2 = S_{n,\bar{X},n}^2$ may have a smaller MSE than S_{n-1}^2 when the variance term is large compared to the bias (see Formula 6). In that case the prefactor $[(n-1)/k]^2$, which is smaller for S_n^2 than for S_{n-1}^2, overcompensates the bias. A large variance of the variance term is especially driven by the

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fourth moment of the population which is the key part of the following theorem stating the situations in which the uncorrected estimator S_n^2 has a smaller MSE than the corrected one S_{n-1}^2.

Theorem 1 .

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample of a random variable X with finite fourth central moment μ_4. Let μ_2 denote the second central moment of X.

1) *If the kurtosis $\mu_4 / \mu_2^2 \geq 1.5$, then*

$$\mathrm{MSE}\left(S_n^2\right) < \mathrm{MSE}\left(S_{n-1}^2\right) \quad (7)$$

holds for all sample sizes n.

2) *For sample sizes $n < 5$, (7) holds for all possible populations.*

3) *For $1 \leq \mu_4 / \mu_2^2 < 1.5$, there exists a sample size n^* such that (7) holds for $n < n^*$ but reverses for $n \geq n^*$.*

Note that in many applications, the first part of the theorem is the relevant one, as the last part, i.e., the case $\mu_4 / \mu_2^2 < 1.5$, corresponds to extremely light-tailed distributions. For the normal distribution, we have the well-known relation $\mu_4 / \mu_2^2 = 3$. Distributions with heavier tails than the normal distribution (e.g., Student's t-distribution, exponential distribution, etc.) satisfy $\mu_4 / \mu_2^2 > 3$. Even for the continuous uniform distribution, a particularly light-tailed case, we have $\mu_4 / \mu_2^2 = 1.8 > 1.5$ (see Table 2 for a compilation of further examples with $\mu_4 / \mu_2^2 > 1.5$). This suggests that, in the majority of cases relevant in practice, the uncorrected variance estimator has a smaller MSE than the corrected one.

Before turning to the proof of Theorem 1, for comprehensiveness, we state a lemma formulating a lower bound of μ_4 given the first central moments of the underlying distribution.

Lemma 2 .

Let X be a random variable with finite fourth central moment μ_4. Let μ_2 and μ_3 denote the second and third central moment of X. Then it holds

$$\mu_4 \geq \left(1 + \frac{\mu_3^2}{\mu_2^3}\right)\mu_2^2 \geq \mu_2^2. \quad (8)$$

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The lemma shows that the theorem applies to all distributions with finite fourth moment. It further follows from the lemma that $\mu_4 = \mu_2^2$ may only be reached for unskewed distributions, i.e., for $\mu_3 = 0,$ which holds, e.g., for symmetric distributions.

A symmetric distribution reaching the boundary $\mu_4 = \mu_2^2$ is given by two equally weighted point masses located at $x = -\delta$ and $x = \delta.$ More general, any linear transformation of a Bernoulli distributed random variable *X* with $P(X = 1) = 1/2 = P(X = 0)$ reaches the boundary. The lemma has been known for a long time going back at least to Pearson (1916) (see also, e.g., Shohat 1929). A short proof is based on the non-negativity of the Hankel determinant:

Proof.

(Lemma 2). Let $Z = X - E[X]$ and denote $m_j = E[Z^j]$ for $j = 0,\ldots,4$, so that $m_j = \mu_j$ for $j = 2,3,4,$ where μ_j are the centered moments of *X*. From the theory of the Hamburger moment problem (see, e.g., Shohat and Tamarkin 1943), it follows that the associated Hankel matrix must be positive semidefinite, and hence its determinant is nonnegative.

$$\begin{vmatrix} m_0 & m_1 & m_2 \\ m_1 & m_2 & m_3 \\ m_2 & m_3 & m_4 \end{vmatrix} = m_2 m_4 - m_3^2 - m_2^3 \geq 0 \rightarrow m_4 \geq \left(1 + \frac{m_3^2}{m_2^3}\right) m_2^2.$$

▪

Note that given the first three moments, there is no upper limit for $\mu_4,$ which is of no harm here, since Part 1 of Theorem 1 holds for all values of $\mu_4 \geq 1.5\mu_2^2$ and thus also for very large values.

We are now prepared to prove Theorem 1:

Proof.

(Theorem 1). We consider Formula (5) for the MSE in the two cases $k = n$ and $k = n - 1$. Equating the resulting expressions, we temporarily treat *n* as a continuous variable and solve for the relationship between kurtosis and sample size that yields equality. By the intermediate value theorem, this allows us to identify regions in the kurtosis–sample-size space where, for integer values of *n*, the mean squared error of S_n^2 is smaller than that of S_{n-1}^2, and vice versa. We obtain the following relationship between kurtosis and sample size for equality:

$$kurt_{\text{equal}} = \left(\frac{3n^2 - 8n + 3}{2n^2 - 3n + 1}\right) := p(n), \quad (9)$$

where $kurt_{\text{equal}}$ refers to the specific kurtosis of the underlying distributions for which both estimators have the same mean squared error for a given integer sample size *n*. By the

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intermediate value theorem, the function $p(n)$ thus separates the regions where, for $n \in \mathbb{N}$, either S_n^2 or S_{n-1}^2 has the smaller mean squared error. It is increasing for $n > 1$ and is bounded by 1.5, i.e., $p(n) < 1.5$ for all $n > 1$.

Now consider the region above the function, e.g., consider a normal distribution (kurtosis $\mu_4 / \mu_2^2 = 3$) and $n = 2$ with $3 > p(2)$. We calculate that $\mathrm{MSE}(S_n^2) < \mathrm{MSE}(S_{n-1}^2)$ using formula (5). We therefore have that $\mathrm{MSE}(S_n^2) < \mathrm{MSE}(S_{n-1}^2)$ holds for all combinations above the function. From $p(n) < 1.5$ for all $n > 1$ follows the first part of Theorem 1.

We also calculate $p(2) = -1/3,\ \ p(3) = 0.6,\ \ p(4) \approx 0.90,\ \ p(5) \approx 1.06.$ Since the kurtosis is bounded from below by 1 ($\mu_4 / \mu_2^2 \geq 1,$ see Lemma 2), we see that for sample sizes below $n = 5$ all population distributions refer to points above the function leading to the second part of Theorem 1.

The third part of the theorem is then a direct consequence of $p(n)$ being increasing and that for every potential kurtosis value $p^* \in [1, 1.5[$ we find a sample size n^* with $p(n^*) < p^*$ and $p(n^* + 1) \geq p^*$. ▪

3.2 Optimal Choice of *k* in $S_{k,M,n}^2$

We now consider an MSE-optimal choice for *k* in estimator S_k^2. Equation (5) and solving

$$\frac{\partial}{\partial k} \mathrm{MSE}(S_k^2) \overset{!}{=} 0$$

for *k* yields to

$$k^{\mathrm{opt}} = \frac{(n-1)\left(\frac{1}{n}\left[\mu_4 - \frac{(n-3)\mu_2^2}{n-1}\right] + \mu_2^2\right)}{\mu_2^2}. \quad (10)$$

Note again that this result holds for all population distributions with $\mu_4 < \infty.$ The optimal parameter *k* is (for fixed μ_2) increasing in $\mu_4.$ The heavier-tailed the distribution is, the larger becomes the optimal value of *k*. For large values of $\mu_4,$ the variance of S_k^2 dominates the MSE and a small factor $\frac{1}{k}$ reduces the contribution of the variance.

For the normal distribution we can substitute $\mu_4 = 3\mu_2^2$ and find $k_{\mathrm{normal}}^{\mathrm{opt}} = n + 1.$ We will illustrate this result in the simulation study in Section 4. Table 2 shows analogous results for other

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distributions. We can see that, in line with the intuition above, k^{opt} increases for increasing kurtosis $\frac{\mu_4}{\mu_2^2}$.

We conclude the section by shortly considering the estimator $S^2_{k,\mu,n}$, i.e., estimating the variance of the population when the population mean is known. For normally distributed populations analogous calculations as above yield to $k^{\text{opt}} = n+2$ as the MSE-optimal factor in this case (see Appendix 8). An interesting result is

$$\text{MSE}\left(S^2_{k=n+2,\mu,n}\right) < \text{MSE}\left(S^2_{k=n+1,\bar{X},n}\right)$$

for all *n* in the case of normally distributed populations (see Appendix 9 for a proof).

3.3 Considerations for Covariances

While the main focus of this paper lies on variance estimation, we briefly consider the analogous problem of estimating covariances. To this end, let $\boldsymbol{X} = (X^{(1)}, \ldots, X^{(d)})$ with $d \geq 2$ denote a random vector and assume that i.i.d. observations $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ are available.

For a fixed pair of indices $j \neq \ell$, we consider the scaled sample covariance estimator

$$S^2_{j\ell} := \frac{1}{k}\sum_{i=1}^{n}(X_i^{(j)} - \bar{X}^{(j)})(X_i^{(\ell)} - \bar{X}^{(\ell)}), \quad (11)$$

which reduces to the usual estimator for $k = n-1$ or $k = n$, respectively.

Here, the MSE-optimal choice of *k* depends on the joint distribution of $X^{(j)}$ and $X^{(\ell)}$, which is made explicit in the following lemma. A detailed proof is provided in Appendix 10.

Lemma 3 .

Let $\sigma_{j\ell} = \text{Cov}(X^{(j)}, X^{(\ell)})$ *and* $\sigma^2_{jj} = \text{Var}(X^{(j)})$, $\sigma^2_{\ell\ell} = \text{Var}(X^{(\ell)})$, *and define*

$$\delta_2^{(j\ell)} = E\left[(X^{(j)} - \mu^{(j)})^2 (X^{(\ell)} - \mu^{(\ell)})^2\right].$$

Then, the MSE-optimal choice of k for the sample covariance estimator $S^2_{j\ell}$ *defined in (11) is given by*

$$k_{\text{opt}} = \frac{1}{n}(n^2 - 2n + 2) + \frac{1}{\sigma^2_{j\ell} n}[(n-1)\delta_2^{(j\ell)} + \sigma^2_{jj}\sigma^2_{\ell\ell}]. \quad (12)$$

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Example 1 .

In the case of multivariate normally distributed variables, the mixed fourth central moment $\delta_2^{(j\ell)}$ *is completely determined by the second moments. In particular, for two jointly Gaussian components* $X^{(j)}$ *and* $X^{(\ell)}$ *with correlation parameter* $\rho = \mathrm{Corr}(X^{(j)}, X^{(\ell)})$ *, it follows from Isserlis' theorem (Isserlis 1918) that*

$$\delta_2^{(j\ell)} = (1+2\rho^2)\sigma_{jj}^2\sigma_{\ell\ell}^2.$$

For completeness, an explicit derivation based on the joint moment generating function is provided in Appendix 11.

Thus, in the special case $\sigma_{jj} = \sigma_{\ell\ell}$ *and* $p = 1$, $\delta_2^{(j\ell)}$ *coincides with the fourth central moment* μ_4 *of the univariate normal distribution. This corresponds to a fully correlated (and hence degenerate) bivariate Gaussian distribution.*

In this setting, the optimal scaling parameter simplifies to $k^{\mathrm{opt}} = n+1$*, which agrees exactly with the optimal choice of the variance estimator obtained for the univariate normal distribution; see Table 2.*

4 Simulation and Variance Estimation in Education

In this section, we illustrate the theoretical results and offer additional insight into the estimators through simulations. We also provide a detailed motivation for the simulation setup and include the corresponding code as a Jupyter notebook in a repository, linked in the Appendix 6. In addition, variance estimation serves as a suitable example for introducing learners to simulation studies. First, learners can see that even relatively simple examples require critical engagement with the literature and the development of a deep understanding. Moreover, variance estimation provides a clear and accessible illustration of the classical trade-off between bias and variance in statistical precision.

4.1 Simulation

In the simulation study, we want to compare bias, variance, mean squared error (MSE), and mean absolute error (MAE) of the variance estimators $S_{k,\bar{X},n}^2$ for $k = n$, $k = n-1$ and for $k = k^{\mathrm{opt}}$. We present the same results for $\sqrt{S_{k,\bar{X},n}^2}$ as an estimator of the standard deviation of the population. We consider several population distributions and sample sizes of $n = 10, 30, 100,$ and report results based on one million Monte Carlo replications. To assess the precision of each of the values of interest, we realize that all of them are expectations of some sort and the simulation result is a sample mean, where the precision may be estimated. Didactically commented computer code for the simulation study is provided in a repository linked in Appendix 6 of this paper.

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Results of the study for $k = n, n-1, k^{opt}$ and different population distributions are shown in Tables 3 and 4. Table 3 summarizes the findings for variance estimators, while Table 4 focuses on standard deviation estimates. Rows refer to combinations of sample size and population distribution, while the columns address the respective value of interest, ranging from the bias over variances and the MSE to the MAE. To make results between rows comparable, all population distributions are scaled to unit variance. Note that the value of k^{opt} depends on the population distribution and is calculated according to Equation (10) and Table 2. By definition, it is optimal for the MSE but may be suboptimal with respect to the MAE. The estimated standard deviations of the simulation results (also see Equation (14) in the next section), i.e., the precision of the results (i.e., standard error of the mean), are reported below the numbers in brackets.

Turning to Table 3, we see that the theoretical results of Section 3 are reflected in the results. For all distributions and all sample sizes the MSE of the $\frac{1}{n}$ estimator is smaller than the MSE of the $\frac{1}{n-1}$ estimator. The positive effect of the apparent bias of the first estimator on the MSE is overcompensated by the effect of its much smaller variance compared with the second estimator. As expected from the theoretical results, the difference of the MSEs of both estimators increases with increasing fourth moments of the distributions (compare to Table 2). Further, we see the same results for the k^{opt} estimator. Its estimations consistently give the smallest MSE for all distributions and all sample sizes. Interestingly and without further theoretical foundation in this paper, the $\frac{1}{n}$ estimator also beats the $\frac{1}{n-1}$ estimator with respect to MAE in all cases.

More surprisingly, the results change when looking at the standard deviation estimates in Table 4. First, we note that in contrast to the variance estimator, the $\frac{1}{n-1}$ estimator for the standard deviation is biased. Thus, the square-root transformation does not only scale the results from Table 3 but affects both the bias and variance of the estimator. We observe that the square-root transformation reduces the variance much more in relation to the bias. Obviously, this is reflected in the MSE. The simulation study indicates that the $\frac{1}{n}$ estimator performs best for all distributions except the uniform distribution. Here the variance is already relatively small compared to the bias for the $\frac{1}{n-1}$ estimator, thus the bias is the dominating element for the MSE of the standard deviation.

Interestingly, considering the MAE instead, we find that the $\frac{1}{n-1}$ estimator yields the smallest values across all distributions and sample sizes. However, due to the square-root in the standard deviation and the absolute value transformation involved in the MAE, analytical treatment for this case is very complicated. Therefore, we do not see a straightforward way to generalize this finding beyond the distributions considered in our simulations.

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4.2 Pedagogical Remarks

The simulation study above can also be used in a learning context by instructors to highlight several topics. First, it serves as an example of what kind of information can be obtained from simulations. Since the variance of the variance estimator is an integral part of the MSE, a simulation study should provide an estimator for it. Moreover, it should be understood how precise the result of the simulation study itself is, that is, one needs an estimator for the variance of the estimator of the variance of the variance estimator.

The explicit formulation of this cascade of variance estimators is not merely a linguistic curiosity. Rather, it can motivate students to clearly distinguish what is being estimated at each level when working with simulation studies.

Furthermore, in contrast to the variance of the sample mean, the variance of the variance estimator depends on higher moments of the population. Variance estimation therefore provides an example in which students can observe the influence of the underlying distribution on the accuracy of an estimator in the simulation results. In this way, they encounter a situation in which assumptions on the population distribution itself, rather than only on its first two moments, become relevant.

In addition, the simulation study serves as an example of how basic statistical concepts can be applied in practice. When, for instance, one is interested in the variance of an estimator $\hat{\theta}$, simulation studies can only provide an estimate of this quantity rather than its true value, since the (outer and inner) expectations in

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}\left[(\hat{\theta} - \mathrm{E}[\hat{\theta}])^2 \right]$$

are approximated by sample means based on *M* Monte Carlo replications of a controlled estimation experiment.

Let $\hat{\theta}_m$, $m = 1, \ldots, M$, denote the estimates obtained from *M* independently generated samples of size *n*. A natural Monte Carlo estimator of $\mathrm{Var}(\hat{\theta})$ is then given by

$$\mathrm{Var}[\hat{\theta}] = \frac{1}{M} \sum_{m=1}^{M} (\hat{\theta}_m - \bar{\hat{\theta}})^2, \quad (13)$$

where $\bar{\hat{\theta}} = \frac{1}{M} \sum_{m=1}^{M} \hat{\theta}_m$. Hence, the Monte Carlo variance estimate itself is a random variable, and repeating the simulation study will generally lead to different outcomes.

Defining now

$$Z_m := (\hat{\theta}_m - \bar{\hat{\theta}})^2, \qquad m = 1, \ldots, M,$$

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the variance estimator in (13) can be written as the sample mean

$$\mathrm{Var}[\hat{\theta}] = \bar{Z} := \frac{1}{M}\sum_{m=1}^{M} Z_m.$$

The key insight for students is that standard tools introduced in elementary statistics courses, such as confidence intervals for the sample mean, are directly applicable to assessing the precision of simulation results. Since $\mathrm{Var}[\hat{\theta}]$ is simply the sample mean of (approximately) independent random variables Z_m, its distributional properties follow from well-known results for $\bar{Z}$.

In particular, for large values of M, $\bar{Z}$ is approximately normally distributed, and its variance can be estimated by

$$\mathrm{Var}[\bar{Z}] = \frac{1}{M}\left(\frac{1}{M}\sum_{m=1}^{M}(Z_m - \bar{Z})^2\right). \quad (14)$$

5 Conclusion

In this paper, we discuss the seemingly simple problem of variance estimation in an i.i.d. sample of a random variable X. We focus on estimators of the form given in Equation (4) and review different choices of k. In a bibliometric survey of statistics textbooks and statistical software, we find that almost every textbook and software uses $k = n-1$, while other choices of k are mentioned rarely. In a theoretically oriented second part, we derive the MSE-optimal choice of k for various distributions. Moreover, we briefly discuss the estimation of the covariance. We illustrate the theoretical results in a simulation study. The simulation study is intended as an introduction to statistical computer simulation for undergraduates and guides the reader step-by-step through the structure of the simulation study. Finally, we argue that the topic of this paper is interesting in a didactic context since it demonstrates to learners the importance of critically approaching literature and building their own deep understanding. Further, we believe variance estimation not only is a simple yet interesting enough example for the classical trade-off between variance and bias as components of precision but also demonstrates the influence of higher moments on sample statistics. We hope this paper helps the readers to gain a deeper understanding of the bias-variance tradeoff and statistics in general.

6 Link to provided computer code

The commented Jupyter Notebook with ready-to-use Python code can be accessed here: https://gitlab.kit.edu/oliver.grothe/variance/

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7 Bias and variance of S_k^2

Here, we provide a derivation of the bias and variance of S_k^2. The basic ideas behind these derivations can be found in Mood, Graybill, and Boes (1974a), which also includes the corresponding results for S_{n-1}^2. Let again $\mu_j = \mathbb{E}[(X-\mu)^j]$ denote the *j*-th central moment of the population and assume $\mu_4 < \infty$. In particular, the variance of the population is $\sigma^2 = \mu_2$. For $1 \le k \le n$ we consider

$$S_k^2 := \frac{1}{k}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Since

$$S_k^2 = \frac{1}{k}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{n-1}{k}S_{n-1}^2, \quad (15)$$

all moments of S_k^2 follow from those of the corrected sample variance S_{n-1}^2.

Bias

We first derive the bias of S_k^2 which follows from its expectation. Using the identity

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2,$$

and noting that $E[(X_i - \mu)^2] = \mu_2$ and $E[(\bar{X} - \mu)^2] = \mu_2 / n$, we obtain

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = n\mu_2 - n\frac{\mu_2}{n} = (n-1)\mu_2.$$

Thus

$$\mathbb{E}[S_{n-1}^2] = \mu_2 = \sigma^2, \qquad \mathbb{E}[S_k^2] = \frac{n-1}{k}\sigma^2,$$

where we have used (15). We see that S_{n-1}^2 is unbiased. The bias for the general case follows as

$$\mathrm{Bias}[S_k^2] = \mathbb{E}[S_k^2] - \sigma^2 = \frac{n-1-k}{k}\sigma^2.$$

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Variance

The trick for the variance is to decompose the estimator into a sum of two components. Let $Y_i := X_i - \mu$ so that $E[Y_i] = 0$, $E[Y_i^2] = \mu_2$ and $E[Y_i^4] = \mu_4$. We set

$$A := \sum_{i=1}^{n} Y_i^2, \qquad \bar{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad B := n\bar{Y}^2$$

and get the decomposition

$$S_{n-1}^2 = \frac{1}{n-1}(A - B).$$

Thus, the variance of S_{n-1}^2 follows from the variances of *A* and *B* and their covariance, which we derive now.

First, we need the variance of *A*. Since the Y_i are i.i.d.,

$$\mathrm{Var}(A) = \sum_{i=1}^{n} \mathrm{Var}(Y_i^2) = n(\mu_4 - \mu_2^2).$$

Calculation of the variance of B ($\mathrm{Var}(B) = \mathbb{E}[B^2] - (E[B])^2$) is a bit more involved. We note that

$$\mathbb{E}[B] = nE[\bar{Y}^2] = \mu_2$$

and

$$\mathbb{E}[B^2] = n^2\mathbb{E}[\bar{Y}^4].$$

That means that we need the fourth moment of $\bar{Y}$:

$$\mathbb{E}[\bar{Y}^4] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right)^4\right] = \frac{1}{n^4}\mathbb{E}\left[\left(\sum_{i=1}^{n} Y_i\right)^4\right].$$

We expand the fourth power and use independence and $E[Y_i] = 0$, which implies that only terms with even total powers in each index survive:

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} Y_i\right)^4\right] = \sum_{i=1}^{n}\mathbb{E}[Y_i^4] + 6\sum_{1\le i<j\le n}\mathbb{E}[Y_i^2 Y_j^2] = n\mu_4 + 3n(n-1)\mu_2^2.$$

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Thus

$$\mathbb{E}[\bar{Y}^4] = \frac{\mu_4}{n^3} + 3\frac{n-1}{n^3}\mu_2^2,$$

which yields

$$\mathbb{E}[B^2] = n^2\mathbb{E}[\bar{Y}^4] = \frac{\mu_4}{n} + 3\frac{n-1}{n}\mu_2^2,$$

and hence

$$\mathrm{Var}(B) = \mathbb{E}[B^2] - (E[B])^2 = \frac{\mu_4 + (2n-3)\mu_2^2}{n}.$$

The last piece is the covariance of A and B. We start with

$$AB = n\left(\sum_{i=1}^{n} Y_i^2\right)\bar{Y}^2 = \frac{1}{n}\left(\sum_{i=1}^{n} Y_i^2\right)\left(\sum_{j=1}^{n} Y_j\right)^2.$$

Thus

$$\mathbb{E}[AB] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{m=1}^{n} Y_i^2 Y_j Y_m\right].$$

Again, because of $E[Y_i] = 0$, the only nonzero contributions occur when each index appears with an even total exponent. There are two situations in which this happens: The n cases where $i = j = m$ lead to a contribution of $E[Y_i^4] = \mu_4$ each and the $n(n-1)$ terms where $j = m \neq i$ which contribute $E[Y_i^2]E[Y_j^2] = \mu_2^2$ each. Hence

$$\mathbb{E}[AB] = \mu_4 + (n-1)\mu_2^2.$$

Since $E[A] = n\mu_2$ and $E[B] = \mu_2$, we get

$$\mathrm{Cov}(A,B) = \mathbb{E}[AB] - \mathbb{E}[A]\mathbb{E}[B] = \mu_4 + (n-1)\mu_2^2 - n\mu_2^2 = \mu_4 - \mu_2^2.$$

We can now compute

$$\mathrm{Var}(A-B) = \mathrm{Var}(A) + \mathrm{Var}(B) - 2\mathrm{Cov}(A,B)$$

by substituting the expressions above and simplifying, which yields

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$$\mathrm{Var}(A-B)=\frac{(n-1)^2}{n}\left[\mu_4-\frac{n-3}{n-1}\mu_2^2\right].$$

Therefore,

$$\mathrm{Var}(S_{n-1}^2)=\frac{1}{(n-1)^2}\mathrm{Var}(A-B)=\frac{1}{n}\left[\mu_4-\frac{n-3}{n-1}\mu_2^2\right].$$

Now, using (15) we get the general case

$$\mathrm{Var}(S_k^2)=\left(\frac{n-1}{k}\right)^2\mathrm{Var}(S_{n-1}^2)=\left(\frac{n-1}{k}\right)^2\frac{1}{n}\left[\mu_4-\frac{n-3}{n-1}\mu_2^2\right].$$

8 k^{opt} for the Population Version of Normal Distributed Variables

When the population mean is known, the variance estimator is given by

$$S_k^{*2}=\frac{1}{k}\sum_{i=1}^{n}\left(X_i-\mu\right)^2.$$

For bias and variance of the estimator follows

$$\mathbb{E}\left[\frac{kS_k^{*2}}{\sigma^2}\right]=n\Leftrightarrow\mathbb{E}\left[S_k^{*2}\right]=\frac{\sigma^2 n}{k}$$

and

$$\mathrm{Var}\left(\frac{kS_k^{*2}}{\sigma^2}\right)=2n\Leftrightarrow\ \mathrm{Var}\left(S_k^{*2}\right)=\frac{2n\sigma^4}{k^2}$$

and therefore

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$$\mathrm{MSE}\left(S_k^{*2}\right)=\mathrm{Var}\left(S_k^{*2}\right)+\mathrm{Bias}^2\left(S_k^{*2}\right)$$
$$=\frac{2n\sigma^4}{k^2}+\left(\frac{\sigma^2 n}{k}-\sigma^2\right)^2$$
$$=\frac{2n\sigma^4}{k^2}+\frac{n^2\sigma^4}{k^2}-\frac{2n\sigma^4}{k}+\sigma^4$$
$$\frac{\partial\mathrm{MSE}\left(S_k^{*2}\right)}{\partial k}\Big|_{k=k^*}=-\frac{4n\sigma^4}{k^{*3}}-\frac{2n^2\sigma^4}{k^{*3}}+\frac{2n\sigma^4}{k^{*2}}\overset{!}{=}0$$
$$\Leftrightarrow -4n\sigma^4-2n^2\sigma^4+2nk^*\sigma^4=0$$
$$\Leftrightarrow -4-2n+2k^*=0$$
$$\Leftrightarrow k^*=n+2$$

9 Comparison of MSEs of $S^2_{k=n+2,\mu,n}$ and $S^2_{k=n+1,\bar{X},n}$

Note that we consider a normally distributed population here. Then $S^2_{k=n+2,\mu,n}$ and $S^2_{k=n+1,\bar{X},n}$ have MSE-optimal factors for the case of known (and used) and unknown population mean μ. Then

$$\Delta\mathrm{MSE}=MSE(S^2_{k=n+2,\mu,n})-MSE(S^2_{k=n+1,\bar{X},n})$$
$$=\left(\frac{2n\sigma^4}{k^{*2}}+\frac{n^2\sigma^4}{k^{*2}}-\frac{2n\sigma^4}{k^*}+\sigma^4\right)-\frac{\sigma^4}{k'^2}\left[2(n-1)+(n-1-k')^2\right]$$
$$=\left[\frac{2n\sigma^4}{(n+2)^2}+\frac{n^2\sigma^4}{(n+2)^2}-\frac{2n\sigma^4}{(n+2)}+\sigma^4\right]-\frac{\sigma^4}{(n+1)^2}\underbrace{\left[2(n-1)+\left(n-1-(n+1)\right)^2\right]}_{=2(n+1)}$$
$$=\frac{2n\sigma^4}{(n+2)^2}+\frac{n^2\sigma^4}{(n+2)^2}-\frac{2n\sigma^4}{(n+2)}+\sigma^4-\frac{2\sigma^4}{n+1}$$
$$=\frac{2n(n+1)\sigma^4+n^2(n+1)\sigma^4-2n(n+2)(n+1)\sigma^4+(n+2)^2(n+1)\sigma^4-2(n+2)^2\sigma^4}{(n+2)^2(n+1)}$$
$$=\left[2n(n+1)+n^2(n+1)-2n(n+2)(n+1)+(n+2)^2(n+1)-2(n+2)^2\right]\frac{\sigma^4}{(n+2)^2(n+1)}$$
$$=\left(2n^2+2n+n^3+n^2-2n^3-6n^2-4n+n^3+5n^2+8n+4-2n^2-8n-8\right)\frac{\sigma^4}{(n+2)^2(n+1)}$$
$$=\underbrace{(-2n-4)}_{<0}\underbrace{\left[\frac{\sigma^4}{(n+2)^2(n+1)}\right]}_{>0}$$
$$\Rightarrow \Delta\mathrm{MSE}<0$$
$$\Rightarrow MSE(S^2_{k=n+2,\mu,n})<MSE(S^2_{k=n+1,\bar{X},n})$$

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10 Proof of Lemma 3

Proof.

(Lemma 3).

We rearrange the non-diagonal elements $S_{j\ell}^2$ as

$$S_{j\ell}^2 = \frac{1}{k}\left(\sum_{i=1}^{n} X_i^{(j)} X_i^{(\ell)} - \frac{1}{n}\sum_{i=1}^{n} X_i^{(j)} \sum_{i=1}^{n} X_i^{(\ell)}\right).$$

Introducing centered variables

$$Y_i^{(j)} := X_i^{(j)} - \mu^{(j)}, \qquad Y_i^{(\ell)} := X_i^{(\ell)} - \mu^{(\ell)},$$

this can be written in the form

$$S_{j\ell}^2 = \frac{1}{k}(A - B), \qquad A := \sum_{i=1}^{n} Y_i^{(j)} Y_i^{(\ell)}, \qquad B := n\bar{Y}^{(j)}\bar{Y}^{(\ell)},$$

where $\bar{Y}^{(j)} = \frac{1}{n}\sum_{i=1}^{n} Y_i^{(j)}$. This decomposition is fully analogous to Appendix 7.

Bias.

Taking expectations and using independence across observations yields

$$E[A] = n\sigma_{j\ell}, \qquad E[B] = \sigma_{j\ell},$$

where $\sigma_{j\ell} = \mathrm{cov}(X^{(j)}, X^{(\ell)})$ denotes the population covariance. Hence,

$$E[S_{j\ell}^2] = \frac{1}{k}(E[A] - E[B]) = \frac{n-1}{k}\sigma_{j\ell}.$$

It follows that $S_{j\ell}^2$ is unbiased for $k = n-1$, and the bias in the general case is given by

$$\mathrm{Bias}[S_{j\ell}^2] = \left(\frac{n-1-k}{k}\right)\sigma_{j\ell}, \qquad j, \ell \in \{1, \ldots, d\}. \quad (16)$$

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Variance.

As in Appendix 7, the variance follows from

$$\mathrm{Var}(S_{j\ell}^2) = \frac{1}{k^2}\mathrm{Var}(A-B) = \frac{1}{k^2}(\mathrm{Var}(A)+\mathrm{Var}(B)-2\mathrm{cov}(A,B)).$$

In contrast to the univariate variance case, the relevant fourth-order quantity is now the mixed central moment

$$\delta_2^{(j\ell)} := E\left[(X^{(j)}-\mu^{(j)})^2(X^{(\ell)}-\mu^{(\ell)})^2\right].$$

Proceeding exactly as in Appendix 7 by expanding the corresponding sums and retaining only index configurations with even total powers, one obtains expressions involving $\delta_2^{(j\ell)}$, $\sigma_{jj}^2\sigma_{\ell\ell}^2$ and $\sigma_{j\ell}^2$ (note that $\sigma_{j\ell}^2 = (\sigma_{j\ell})^2$ is the square of the covariance in our notation).

Collecting terms yields

$$\mathrm{Var}(S_{j\ell}^2) = \frac{n-1}{nk^2}[(n-1)\delta_2^{(j\ell)} + \sigma_{jj}^2\sigma_{\ell\ell}^2 - (n-2)\sigma_{j\ell}^2], \quad (17)$$

which completes the derivation.

Setting

$$\frac{\partial}{\partial k}\mathrm{MSE}(S_{j\ell}^2) = \frac{\partial}{\partial k}(\mathrm{Var}(S_{j\ell}^2) + \mathrm{Bias}[S_{j\ell}^2]^2) \stackrel{!}{=} 0,$$

and substituting the expressions for $\mathrm{Var}[S_{j\ell}^2]$ and $\mathrm{Bias}[S_{j\ell}^2]$ derived above, we obtain after straightforward algebra

$$\frac{\partial}{\partial k}\mathrm{MSE}(S_{j\ell}^2) = -\frac{2(n-1)}{nk^3}[(n-1)\delta_2^{(j\ell)} + \sigma_{jj}^2\sigma_{\ell\ell}^2 - (n-2)\sigma_{j\ell}^2 + n\sigma_{j\ell}^2(n-1-k)].$$

Since the prefactor is nonzero for $n \neq 1$, solving the first-order condition yields

$$k_{\mathrm{opt}} = \frac{1}{n}(n^2-2n+2) + \frac{1}{n\sigma_{j\ell}^2}[(n-1)\delta_2^{(j\ell)} + \sigma_{jj}^2\sigma_{\ell\ell}^2]. \quad (18)$$

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Here, $\delta_2^{(j\ell)}$ denotes the mixed fourth central moment defined above, which may be obtained as the second derivative of the joint moment generating function $M_{(X^{(j)},X^{(\ell)})}(t_1,t_2)$; see Section 3.9 in Mittelhammer 2013a.

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11 Details on the Mixed Fourth Central Moment $\delta_2^{(j\ell)}$ for Bivariate Normal

Recall that $\delta_2^{(j\ell)} = E[(X^{(j)}-\mu^{(j)})^2(X^{(\ell)}-\mu^{(\ell)})^2]$ represents the second moment of the variance or second cross moment, i.e., the second derivative of the joint moment generating function $M_{(X_j,X_l)}(t_1,t_2)$. For the case of multivariate normal distributed variables, it results in

$$
\begin{aligned}
\delta_2^{(j\ell)} &= \frac{\partial^4}{\partial t_j^2 \partial t_l^2} M_{(X_j,X_l)}(t_j,t_l) \\
&= \sigma_{\ell\ell}\rho\sigma_{jj}[(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)})+\sigma_{\ell\ell}\rho\sigma_{jj}]M_{(X_j,X_l)}(t_j,t_l) \\
&+(\sigma_{\ell\ell}^2\rho^2\sigma_{jj}^2+\sigma_{\ell\ell}^2\sigma_{jj}^2)M_{(X_j,X_l)}(t_j,t_l) \\
&+\sigma_{\ell\ell}^2(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)})(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)})M_{(X_j,X_l)}(t_j,t_l) \\
&+3\sigma_{\ell\ell}\rho\sigma_{jj}(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)})M_{(X_j,X_l)}(t_j,t_l) \\
&+\sigma_{jj}^2(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})M_{(X_j,X_l)}(t_j,t_l) \\
&+(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})(t_l\sigma_{\ell\ell}^2+\sigma_{\ell\ell}\rho\sigma_{jj}t_j+\mu^{(\ell)})(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)}) \\
&\cdot(t_l\sigma_{\ell\ell}\rho\sigma_{jj}+\sigma_{jj}^2t_j+\mu^{(j)})M_{(X_j,X_l)}(t_j,t_l)
\end{aligned}
$$

with $M_{(X_j,X_l)} = exp[t_1\mu^{(j)}+t_1\mu^{(\ell)}-\frac{1}{2}(t_1^2\sigma_{jj}^2+t_2^2\sigma_{\ell\ell}^2+2\rho t_1t_2\sigma_{jj}\sigma_{\ell\ell})]$ (see Section 3.9, Mittelhammer 2013a).

Then setting $M_{(X_j,X_l)}(0,0)=1$ and assuming $t_j=t_l=\mu^{(j)}=\mu^{(\ell)}=0$, we obtain

$$
\begin{aligned}
\delta_2^{(jl)} &= \sigma_{jj}^2\sigma_{\ell\ell}^2\rho^2+\sigma_{jj}^2\sigma_{\ell\ell}^2\rho^2+\sigma_{jj}^2\sigma_{\ell\ell}^2+0+0+0 \\
&= (2\rho^2+1)\sigma_{jj}^2\sigma_{\ell\ell}^2.
\end{aligned}
$$

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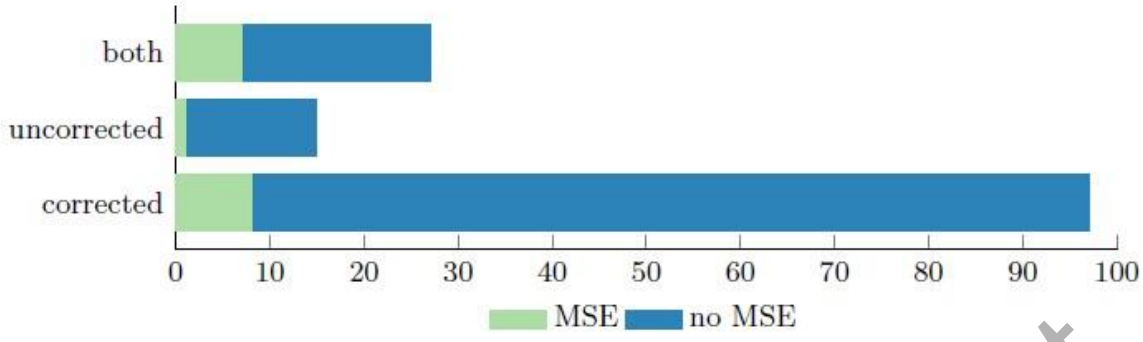

Figure 1: The number of considered books, which introduced the corrected, uncorrected or both estimators.

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Table 1: Packages and functions to calculate the variance of data `X` per statistics software.

Software	Package/Class	Corrected Function	Uncorrected Function	Standard Setting
EViews	base	`@var(X)`	`@varp(X)`	Corrected
Excel	base	`VAR.S(X)`	`VAR.P(X)`	Corrected / Uncorrected
Java	Variance	Variance(X)	-	Corrected
KNIME	Statistics Node	no code	-	Corrected
Limdep	base	Var(X)		Corrected
Maple	Statistics	`Variance(X)`	-	Corrected
Matlab	Statistics and ML Toolbox	`var(X)`	`var(X,1)`	Corrected
Orange	Aggregate Columns	no code	-	Corrected
PPSP	base	VARIANCE(X)	-	Corrected
Python	numpy	`numpy.var(X,ddof=1)`	`numpy.var(X)`	Uncorrected
R	base	`var(X)`	-	Corrected
RapidMiner	base	no code	-	Corrected
SAS	base	`var(X)`	-	Corrected
SPSS	Descriptive Statistics	no code	-	Corrected
SQL	base	`VAR(X)`	`VARP()`	Corrected
STATISTICA	base	no code	-	Corrected
Stata	base	`summarize X / var X`	-	Corrected
WEKA	base	no code	-	Corrected
Wolfram	base	`VARIANCE[X]`	-	Corrected
gretl	base	`var()`	-	Corrected

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Table 2: Factor k^{opt} of the MSE-minimal estimator of the variance for estimators of type S_k^2 for populations of different distributions. The parameters of the distributions are: (1) uniform on $[a,b]$; (2) $v>0$: degrees of freedom; (3) $\theta>0$: expectation; (4) $\lambda>0$: expectation. For convenience, the fourth central moment μ_4 is reported as a multiple of μ_2^2. In all cases, the corresponding kurtosis μ_4/μ_2^2 exceeds 1.5. Consequently, by Theorem 1, the uncorrected sample variance has a smaller mean squared error than the corrected one for all distributions listed here, and thus we have $\text{MSE}(S_{k^{\text{opt}}}^2)<\text{MSE}(S_n^2)<\text{MSE}(S_{n-1}^2)$.

Distribution	μ_2	μ_4	k^{opt}
Normal	σ^2	$3\mu_2^2$	$n+1$
Uniform (1)	$(b-a)^2/12$	$9/5\mu_2^2$	$n+6/(5n)-1/5$
Student-t (2)	$v/(v-2)$	$3v^2/[(v-2)(v-4)]=3\frac{v-2}{v-4}\mu_2^2$	$6(n-1)/[n(v-4)]+n+1$
χ^2 (2)	$2v$	$12v(v+4)=3\frac{(v+4)}{v}\mu_2^2$	$12(n-1)/(nv)+n+1$
Exponential (3)	θ^2	$9\mu_2^2$	$(n^2+7n-6)/n$
Poisson (4)	λ	$\lambda(3\lambda+1)=(3+\frac{1}{\lambda})\mu_2^2$	$(n-1)/(n\lambda)+n+1$

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Table 3: Simulation results for S_k^2. In each row, the estimator that performs best according to the respective performance measure is highlighted in bold.

		Bias			**Var**			**MSE**			**MAE**		
Distribution	*n*	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$
Normal	10	-0.1002	-0.0002	**-0.1820**	0.1800	0.2222	**0.1487**	0.1900	0.2222	**0.1818**	**0.3537**	0.3691	0.3559
		(0.0004)	(0.0005)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0003)	(0.0004)	(0.0002)	(0.0003)	(0.0003)	(0.0002)
	30	-0.0330	0.0004	**-0.0641**	0.0644	0.0689	**0.0603**	0.0655	0.0689	**0.0644**	**0.2052**	0.2082	0.2057
		(0.0003)	(0.0003)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0002)	(0.0001)
	100	-0.0098	0.0002	**-0.0196**	0.0199	0.0203	**0.0195**	0.0200	0.0203	**0.0198**	**0.1129**	0.1134	0.1129
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
Uniform	10	**-0.1000**	-0.0000	-0.0928	**0.0829**	0.1024	0.0842	0.0929	0.1024	**0.0928**	0.2475	0.2571	**0.2472**
		(0.0003)	(0.0003)	(0.0003)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0002)	(0.0002)
	30	**-0.0335**	-0.0002	-0.0283	**0.0271**	0.0290	0.0274	0.0282	0.0290	**0.0282**	0.1348	0.1361	**0.1347**
		(0.0002)	(0.0002)	(0.0002)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
	100	**-0.010**	-0.000	-0.008	**0.0081**	0.0082	0.0081	0.0082	0.0082	**0.0081**	0.0722	0.0723	**0.0721**

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		Bias			**Var**			**MSE**			**MAE**		
Distribution	*n*	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\text{opt}}^2$
		1	1	2									
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
Student-t (6)	10	-0.1005	-0.0005	**-0.3434**	0.4099	0.5061	**0.2184**	0.4200	0.5061	**0.3363**	**0.4581**	0.4750	0.4832
		(0.0006)	(0.0007)	(0.0005)	(0.0047)	(0.0058)	(0.0025)	(0.0046)	(0.0058)	(0.0024)	(0.0005)	(0.0005)	(0.0003)
	30	-0.0337	-0.0004	**-0.1449**	0.1597	0.1709	**0.1251**	0.1608	0.1709	**0.1461**	**0.2864**	0.2892	0.2961
		(0.0004)	(0.0004)	(0.0004)	(0.0036)	(0.0039)	(0.0028)	(0.0036)	(0.0039)	(0.0028)	(0.0003)	(0.0003)	(0.0002)
	100	-0.0101	-0.0001	**-0.0479**	0.0494	0.0504	**0.0457**	0.0495	0.0504	**0.0480**	**0.1653**	0.1657	0.1677
		(0.0002)	(0.0002)	(0.0002)	(0.0008)	(0.0008)	(0.0007)	(0.0008)	(0.0008)	(0.0007)	(0.0001)	(0.0002)	(0.0001)
$\chi^2(6)$	10	-0.0999	0.0001	**-0.2968**	0.3419	0.4221	**0.2087**	0.3519	0.4221	**0.2967**	**0.4541**	0.4728	0.4628
		(0.0006)	(0.0006)	(0.0005)	(0.0011)	(0.0013)	(0.0007)	(0.0010)	(0.0013)	(0.0005)	(0.0004)	(0.0004)	(0.0003)
	30	-0.0331	0.0002	**-0.1192**	0.1270	0.1359	**0.1054**	0.1281	0.1359	**0.1196**	**0.2791**	0.2827	0.2820
		(0.0004)	(0.0004)	(0.0003)	(0.0003)	(0.0003)	(0.0002)	(0.0003)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
	10	-	0.000	**-**	0.039	0.040	**0.037**	0.039	0.040	**0.038**	**0.157**	0.157	0.157

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		Bias			**Var**			**MSE**			**MAE**		
Distribution	*n*	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$
	0	0.0097	3	**0.0384**	4	2	**1**	5	2	**6**	**2**	8	8
		(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Exponential	10	-0.0999	0.0001	**-0.4512**	0.6666	0.8230	**0.2479**	0.6766	0.8230	**0.4514**	**0.5893**	0.6171	0.5915
		(0.0008)	(0.0009)	(0.0005)	(0.0030)	(0.0037)	(0.0011)	(0.0029)	(0.0037)	(0.0009)	(0.0006)	(0.0007)	(0.0003)
	30	-0.0335	-0.0002	**-0.2121**	0.2516	0.2693	**0.1672**	0.2528	0.2693	**0.2122**	**0.3789**	0.3844	0.3830
		(0.0005)	(0.0005)	(0.0004)	(0.0007)	(0.0008)	(0.0005)	(0.0007)	(0.0008)	(0.0004)	(0.0003)	(0.0003)	(0.0003)
	100	-0.0098	0.0002	**-0.0740**	0.0790	0.0806	**0.0690**	0.0791	0.0806	**0.0745**	**0.2188**	0.2197	0.2201
		(0.0003)	(0.0003)	(0.0003)	(0.0002)	(0.0002)	(0.0001)	(0.0002)	(0.0002)	(0.0001)	(0.0002)	(0.0002)	(0.0002)
Poisson	10	-0.0998	0.0002	**-0.2435**	0.2621	0.3236	**0.1851**	0.2721	0.3236	**0.2444**	**0.4094**	0.4267	0.4174
		(0.0005)	(0.0006)	(0.0004)	(0.0007)	(0.0008)	(0.0005)	(0.0006)	(0.0008)	(0.0004)	(0.0003)	(0.0004)	(0.0003)
	30	-0.0329	0.0004	**-0.0924**	0.0956	0.1023	**0.0842**	0.0967	0.1023	**0.0927**	**0.2461**	0.2494	0.2477
		(0.0003)	(0.0003)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0002)	(0.0002)	(0.0002)

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		Bias			**Var**			**MSE**			**MAE**		
Distribution	*n*	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$	S_n^2	S_{n-1}^2	$S_{n,\mathrm{opt}}^2$
	100	-0.0101	-0.0001	**-0.0294**	0.0296	0.0302	**0.0285**	0.0297	0.0302	**0.0293**	**0.1372**	0.1378	0.1376
		(0.0002)	(0.0002)	(0.0002)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)

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Table 4: Simulation results for $\sqrt{S_k^2}$. In each row, the estimator that performs best according to the respective performance measure is highlighted in bold.

		Bias			**Var**			**MSE**			**MAE**		
Distribution	***n***	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\text{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\text{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\text{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\text{opt}}^2}$
Normal	10	-0.0773	-0.0274	**-0.1203**	0.0485	0.0539	**0.0441**	**0.0545**	0.0547	0.0586	0.1892	**0.1878**	0.1979
		(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
	30	-0.0250	-0.0084	**-0.0409**	0.0165	0.0171	**0.0160**	**0.0171**	0.0171	0.0176	0.1049	**0.1047**	0.1068
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
	100	-0.0074	-0.0024	**-0.0124**	0.0050	0.0051	**0.0050**	**0.0051**	0.0051	0.0051	0.0568	**0.0568**	0.0571
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Uniform	10	**-0.0643**	-0.0137	-0.0605	**0.0244**	0.0271	0.0246	0.0285	**0.0273**	0.0283	0.1328	**0.1313**	0.1322
		(0.0002)	(0.0002)	(0.0002)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
	30	**-0.0205**	-0.0038	-0.0179	**0.0071**	0.0074	0.0072	0.0075	**0.0074**	0.0075	0.0689	**0.0685**	0.0687
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
	10	**-**	-	-	**0.002**	0.002	0.002	0.002	**0.002**	0.002	0.036	**0.036**	0.036

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		Bias			**Var**			**MSE**			**MAE**		
Distribution	***n***	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$
	0	**0.0061**	0.0011	0.0052	**0**	1	0	1	**1**	1	3	**2**	3
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Student-t (6)	10	-0.0955	-0.0466	**-0.2273**	0.0815	0.0905	**0.0595**	**0.0906**	0.0927	0.1111	0.2407	**0.2372**	0.2853
		(0.0003)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0002)	(0.0002)	(0.0002)
	30	-0.0340	-0.0175	**-0.0913**	0.0331	0.0342	**0.0293**	**0.0342**	0.0345	0.0376	0.1445	**0.1435**	0.1578
		(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
	100	-0.0107	-0.0057	**-0.0298**	0.0112	0.0113	**0.0108**	**0.0113**	0.0113	0.0116	0.0826	**0.0823**	0.0854
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
$\chi^2(6)$	10	-0.0937	-0.0446	**-0.1989**	0.0787	0.0874	**0.0615**	**0.0874**	0.0894	0.1010	0.2406	**0.2382**	0.2696
		(0.0003)	(0.0003)	(0.0002)	(0.0001)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0002)	(0.0002)
	30	-0.0323	-0.0157	**-0.0764**	0.0304	0.0314	**0.0277**	**0.0314**	0.0317	0.0335	0.1420	**0.1414**	0.1497
		(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000	(0.000

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		Bias			Var			MSE			MAE		
Distribution	***n***	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$
		2)	2)	2)	0)	1)	0)	0)	0)	0)	1)	1)	1)
	100	-0.0097	-0.0048	**-0.0242**	0.0097	0.0098	**0.0094**	**0.0098**	0.0098	0.0100	0.0790	**0.0789**	0.0804
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
Exponential	10	-0.1236	-0.0762	**-0.3156**	0.1320	0.1467	**0.0805**	**0.1473**	0.1525	0.1801	0.3129	**0.3115**	0.3731
		(0.0004)	(0.0004)	(0.0003)	(0.0003)	(0.0003)	(0.0002)	(0.0002)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
	30	-0.0457	-0.0294	**-0.1384**	0.0559	0.0578	**0.0455**	**0.0580**	0.0587	0.0647	0.1927	**0.1923**	0.2127
		(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
	100	-0.0144	-0.0094	**-0.0469**	0.0188	0.0190	**0.0176**	**0.0190**	0.0191	0.0198	0.1098	**0.1097**	0.1141
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
Poisson	10	-0.0858	-0.0363	**-0.1619**	0.0644	0.0715	**0.0541**	**0.0717**	0.0728	0.0803	0.2175	**0.2155**	0.2375
		(0.0003)	(0.0003)	(0.0002)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0002)	(0.0002)
	30	-0.028	-0.012	**-0.059**	0.0236	0.0244	**0.0221**	**0.0244**	0.0246	0.0256	0.1255	**0.1250**	0.1300

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		Bias			**Var**			**MSE**			**MAE**		
Distribution	***n***	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$	$\sqrt{S_n^2}$	$\sqrt{S_{n-1}^2}$	$\sqrt{S_{n,\mathrm{opt}}^2}$
		7	1	**0**									
		(0.0002)	(0.0002)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)
	100	-0.0088	-0.0038	**-0.0185**	0.0074	0.0074	**0.0072**	**0.0074**	0.0075	0.0076	0.0690	**0.0689**	0.0698
		(0.0001)	(0.0001)	(0.0001)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0001)

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